Votes Without Neighborhoods: Housing Prices and the Geography of Conservative Voting in Seoul

Abstract. Higher apartment prices are associated with greater conservative support across Seoul's districts. Does the association also run between neighborhoods within districts, including outside Gangnam, Seocho and Songpa? Korean neighborhood returns answer only in part: election-day ballots are counted where voters live, but early ballots, a third to a half of the total, are counted where they were cast or reported only for larger units. Among election-day voters, the within-district gradient is point-identified and positive in every election from 2022, as in 2012, also outside the three southeastern districts; in 2017 it is close to zero. For all residents, we bound it over every ballot allocation consistent with official returns, electoral rolls and statutory early-voting zones, then add explicit limits on early voters' movement and vote composition. The returns alone establish no direction in any election with nationwide early voting. Under the main assumptions the gradient among all residents is positive in 2026, also outside the three districts if out-of-zone voters differ from their local station by up to 10 points but not 20; it is narrowly positive in 2024 and undetermined in 2022 and 2025. The finer 2026 zones narrow these ranges. Under the main assumptions, reporting out-of-zone ballots by registered neighborhood would remove 27–57% of the remaining range.

Keywords: electoral geography; housing politics; Gangnam; early voting; election administration; South Korea

1. Introduction

When Oh Se-hoon was re-elected mayor of Seoul in June 2026, maps of his vote by neighborhood circulated within days. They showed what Korean political geography has described for two decades: the conservative vote rising with apartment prices, from the high-rise districts south of the Han River to the older neighborhoods of the north and west. A polling firm's research note put a number on it: across Seoul's 424 administrative neighborhoods (*dong*, 행정동), the correlation between apartment prices and Oh's share was 0.84, and 0.71 within districts (Lee 2026). Such figures feed a long-running debate about whether housing has become a political cleavage in Korea (Son 2008, 2010; Lee and Kwon 2019; Han 2024).

These figures raise a question about the spatial reach of Seoul's housing politics. District-level evidence cannot show whether the conservatism of expensive areas is concentrated in Gangnam and its neighbors, whose residents have built and defended an exclusive identity (Bae and Joo 2020), or whether it also separates neighborhoods of the same district elsewhere in the city. We know of no account that confines the price–vote association to three districts, and place identities can form at the neighborhood scale anywhere, so our question concerns the reach of the association rather than its mechanism. Three spatial claims are involved: that districts (*gu*, 구) with higher apartment prices give more support to conservative candidates; that the same holds between neighborhoods within districts, the scale of housing markets and redevelopment politics; and that it holds beyond the wealthy southeastern districts of Gangnam, Seocho and Songpa. A district-level association need not recur within districts, and a within-district slope may rest on a few neighborhoods.

Korean neighborhood returns cannot settle these claims directly because of how ballots are attributed. Since nationwide early voting began in 2014, a large and growing share of ballots has been cast before election day. In recent Seoul elections, 22–35% of all ballots were early ballots cast inside the voter's designated early-voting zone (the district in presidential elections; Section 3.1). These are counted for the neighborhood of the station where they were cast, which need not be the voter's own. A further 12–15% were early ballots cast outside the zone, or at-home and overseas ballots, and are reported only for the district or constituency. A neighborhood's published return therefore includes some of its neighbors' votes and omits some of its residents'. Only election-day ballots, counted in the voter's own voting district, are attributed exactly. The problem has grown with the partisan gap between early and election-day voters. The conservative candidate's share among in-zone early voters was on average 4 points lower than among election-day voters in the same district in 2017, and 10, 12, 28 and 26 points lower in 2022, 2024, 2025 and 2026 (Online Appendix Table C8). Korean early voters differ in composition from election-day voters (Ka 2021), and convenience voting can change who votes when (Gronke et al. 2008; Kaplan and Yuan 2020). The gap widened as claims spread after 2020 that early-vote counts had been manipulated, claims the election commission rejected (National Election Commission 2022b); the returns document the gap but not its causes. With gaps of this size, assigning an early ballot to the wrong neighborhood, or dropping a resident's out-of-zone ballot, can create or remove a neighborhood difference.

We compute the range of each quantity over all ballot allocations consistent with the published counts and then examine how the range narrows as explicit assumptions are added. Across districts, higher apartment prices are associated with greater conservative support in all six elections without any allocation assumption, although in 2012 and 2017 the association rests largely on the three southeastern districts. Within districts, the association is point-identified for election-day voters: their conservative share rises with neighborhood prices in 2012 and in every election from 2022, including outside Gangnam, Seocho and Songpa, and is nearly flat in 2017. Whether the same holds for all residents depends on how early voters are allocated. Under our main assumptions, residents' early votes may depart from their neighborhood's election-day vote by about as much as the published station returns do, plus two points. With these assumptions a positive slope among all residents is identified in 2026, including outside the three districts, where a neighborhood with apartment prices twice those of another in the same district has a fitted conservative share 2.0 to 9.6 points higher. The 2024 slope is narrowly positive but fragile, and the 2022 and 2025 slopes are undetermined. A further restriction, that early-vote departures are no more aligned with prices than observed station patterns, would make the slope positive for all of Seoul in those years as well, but it cannot be verified. Part of the difference between elections in how clearly the relation can be seen is administrative. Under weaker assumptions, applying the 2026 zones to the same returns decides whether a positive slope is identified; under the main assumptions, the zones narrow the range by about a quarter. A clearer result in 2026 is therefore not by itself evidence that Seoul's housing divide deepened.

The paper contributes to the study of housing politics and elections in three ways. It establishes the spatial reach of the price–vote association in Seoul: the association differs across spatial scales, and within districts it extends beyond Gangnam, Seocho and Songpa, exactly for election-day voters and, in 2026 and under stated assumptions, for all residents. It shows that whose votes are observed matters for that conclusion. Because early and election-day voters now differ sharply in partisanship, extending a pattern observed among election-day voters to all residents is an inference about a politically distinct group of voters, and the paper makes the required assumptions explicit. And it shows how counting and reporting rules limit what can be learned about neighborhood political differences, so that an election revealing a clearer within-district relation need not be one with a deeper housing divide; reporting out-of-zone ballots by registered neighborhood would narrow what remains unknown. Methodologically, the paper treats a familiar attribution problem in electoral geography as a partial-identification problem. Unlike the modifiable areal unit problem, which concerns the size and boundaries of spatial units (Lee et al. 2025), this problem arises within fixed boundaries, because the votes reported for an area are not necessarily those of its residents. Postal ballots in Germany and declaration votes in Australia are likewise reported apart from polling-place results (Online Appendix B.5), and United States precinct datasets must allocate absentee ballots (Amos et al. 2017; Baltz et al. 2022).

2. Housing and conservative voting in Seoul

Housing is a central axis of Korean political conflict. Son's (2008, 2010) work on Seoul and the capital region found that homeownership, apartment residence and housing values tracked conservative voting across neighborhoods, a "real-estate class society". Survey research relates homeownership to vote choice (Lee and Kwon 2019; Shin 2022) and subjective class position to conservative support (Han 2026), and studies of administrative units relate housing values to conservative support across municipalities, districts and constituencies (Oh and Park 2013; Han 2024; Song and Kang 2025). In the transactions we use, the median neighborhood's price per square meter roughly doubled between the twelve months before the 2017 and the 2022 presidential elections, and the conservative opposition won the 2021 Seoul mayoral by-election and the 2022 presidential election. Comparative work links housing wealth and local housing markets to political preferences and populist support (Ansell 2014, 2019; Larsen et al. 2019; Adler and Ansell 2020; Ansell et al. 2022), and in the United States buying a home increases turnout in local elections (Hall and Yoder 2022).

Existing studies address different spatial scales. Han (2024) relates conservative support to average apartment prices across districts, treating housing wealth as a regional characteristic. Bae and Joo (2020) explain Gangnam's conservatism through the construction of place identity: residents' efforts to preserve the area's exclusivity and transmit it to their children. Their account raises the question of how much a city-wide price gradient depends on this distinctive area. Dong-level studies examine finer spatial differences, relating price levels or price changes to electoral outcomes (Kim and Kang 2022; Yun and Jun 2024; Lee 2026). Such studies address related but distinct quantities; an association involving price changes does not directly test the price-level gradient examined here. Substantial variation within districts makes the latter question relevant: in our 2026 transaction data, median apartment prices per square meter differ by a median of 24% between adjacent neighborhoods in the same district, and 39% of the variance of log neighborhood prices lies within districts.

Owners' economic interests (Fischel 2001; Marble and Nall 2021), residential sorting (Rodden 2019) and local context could each produce these associations, and a gradient outside Gangnam would not distinguish among them, since local identities can develop anywhere and housing interests may depend on local conditions (Hankinson 2018). We therefore examine the spatial reach of the pattern rather than its mechanism. The primary estimand is the *gradient*, the within-district slope of the conservative share on log price. Because it gives most weight to neighborhoods far from their district's mean log price, we also report how much it depends on the most expensive neighborhoods (Section 4.5). As secondary summaries we use *contrasts* between each district's higher- and lower-price neighborhoods (above versus at or below the median, and top versus bottom third), averaged over districts with weights proportional to their number of neighborhoods with price data. Slopes and contrasts are different estimands with different weights and units, and in these data the sign of the contrasts is less stably identified than the sign of the slope.

3. Electoral reporting, data and identification

3.1 Electoral reporting rules

Korea reports election results by voting district (투표구) and therefore by administrative neighborhood, a unit with about 19,500 electors in Seoul. Box 1 and Figure 1 summarize where each kind of ballot is counted. Election-day ballots are counted in the voter's own voting district. Since the April 2013 by-elections and nationwide since 2014, any voter can also vote early at any station in the country. In-zone early voters (관내사전투표) cast a ballot within their designated zone, placing it directly in the station's ballot box; it is then counted for the station's neighborhood. The boundary that matters is this statutory zone, which coincides with the district only in some elections; we therefore write in-zone and out-of-zone rather than the literal in-district and out-of-district. Out-of-zone early voters (관외사전투표) seal their ballot in an envelope that is mailed to their district commission and counted only for the district, as are at-home (거소) and overseas ballots. Before 2014, voters who could not vote on election day registered as absentees (부재자) and their ballots were likewise reported only for the district; only 3.4% of Seoul ballots were pooled in 2012.

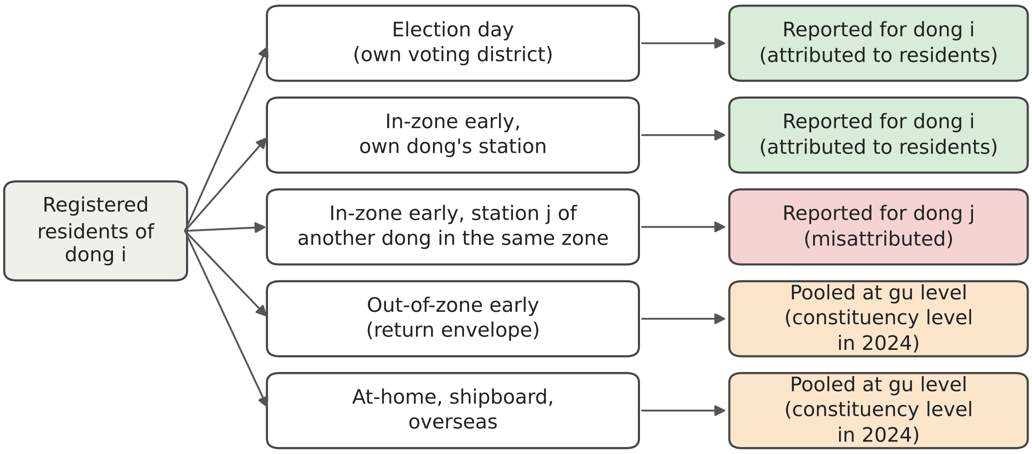

Registered residents of dong i
Election day (own voting district)
In-zone early, own dong's station
In-zone early, station j of another dong in the same zone
Out-of-zone early (return envelope)
At-home, shipboard, overseas
Reported for dong i (attributed to residents)
Reported for dong i (attributed to residents)
Reported for dong j (misattributed)
Pooled at gu level (constituency level in 2024)
Pooled at gu level (constituency level in 2024)

Figure 1: Where a Seoul resident's ballot is reported, by voting mode, since 2014. The zone is the district; in National Assembly elections, the constituency where a district holds two or more; in local elections, the district-council constituency.

Box 1. Where Korean ballots are reported

Legal sources: Public Official Election Act (공직선거법; Act; Korea n.d.) and Regulations on Public Official Elections (공직선거관리규칙; Regulations; National Election Commission n.d.), in the versions in force at each election.

Ballot	Rule	Where reported
Election day	Counted and published by voting district (Act, Art. 178)	Voting district, hence neighborhood
In-zone early	No envelope inside the zone (Act, Art. 158(5); zone: Regulations, Arts. 86(3), 133(3)); each station's box is counted separately	Station's neighborhood
Out-of-zone early	Envelope mailed to the voter's district commission (Act, Arts. 158, 176)	District or constituency
At-home, shipboard, overseas	Registered in advance; mailed to the district commission	District or constituency
Election-day roll	Printed after early voting, early voters marked (Regulations, Art. 16-2)	Residents removed from the roll

The early-voting zone is decisive for our question, because a voter can be misattributed to another neighborhood only inside it. In presidential elections the zone is the whole district, about 17 neighborhoods; in National Assembly elections it is the constituency where a district holds two or more (Regulations, Art. 86(3)). In the 2026 local elections it was the district-council constituency (Regulations, Art. 133(3); National Election Commission 2022a): 155 constituencies in Seoul, with 2.8 neighborhoods on average. The council returns are consistent with this restriction: at all 320 early-voting stations in contested constituencies, the number of early voters in the council contest equals the number in the mayoral contest. Two features of the published data make inference possible. The election-day roll is printed after early voting with early voters marked, so the difference between each neighborhood's electorate and its election-day roll counts the residents removed from that roll: those who voted early or were registered to vote at home or abroad, some of whom cast no counted ballot. And pooled-ballot totals are published by candidate for each counting jurisdiction. With each station's count, these numbers constrain how the unattributed ballots can have been distributed.

3.2 Data

We study the presidential elections of 2012, 2017, 2022 and 2025, the 2024 National Assembly election and the 2026 election of the mayor of Seoul (Table 1). Returns and electorates are the National Election Commission's official counting-unit files and electorate records (National Election Commission 2026); the 2026 zones come from its official table of council constituencies and neighborhoods. The outcome is the share of the main conservative candidate: Park Geun-hye (2012), Hong Joon-pyo (2017), Yoon Suk Yeol (2022), the People Power Party's constituency candidates (2024), Kim Moon-soo (2025) and Oh Se-hoon (2026). In 2017 the conservative vote was split; combining Hong with Yoo Seung-min raises the conventional within-district slope from 0.014 to 0.041 but leaves it undetermined with the returns alone and under minimal movement ($[-0.500, 0.556]$ and $[-0.154, 0.225]$; Online Appendix Table C24). Apartment prices are medians of sale prices per square meter recorded by the Ministry of Land, Infrastructure and Transport (2026) in the twelve months before each election, geocoded to neighborhoods; they measure the price level of a neighborhood's apartment stock, not ownership or wealth. Mobility data come from Seoul Living Mobility (Seoul Metropolitan Government and KT 2026), which estimates trips from mobile-phone signals and is available from 2022. Online Appendix B describes the sources and their construction.

Table 1. Elections, reporting units and ballot coverage

Election	Pooled ballots reported for	In-zone early zone	Dongs	Dongs with price	Pooled share	In-zone early share	Out-of-zone share of early ballots	Stations with forced movement
2012 pres.	25 gu	–	423	414	3.4%	–	–	0
2017 pres.	25 gu	25 gu	424	421	12.8%	21.7%	34.6%	5
2022 pres.	25 gu	25 gu	425	420	13.5%	35.3%	26.1%	20
2024 leg.	48 constituencies	48 constituencies	425	418	12.9%	34.8%	26.1%	20
2025 pres.	25 gu	25 gu	426	420	14.7%	29.2%	31.7%	13
2026 mayor	25 gu	155 council constituencies	427	423	12.1%	25.6%	31.8%	8

Notes. Shares are of all ballots cast in Seoul. Pooled ballots: out-of-zone early, at-home, shipboard and overseas. "Out-of-zone share of early ballots" is P/(P+K) over Seoul. Forced movement: stations whose in-zone early ballots exceed their dong's removed residents ($r_j > 0$). 2012 had no in-zone early voting.

3.3 Identification

An illustrative district.

Consider a district with two neighborhoods, a and b, each with 1,000 residents who cast a ballot. Of these, 600 in each neighborhood voted on election day, and candidate X received 330 ballots in a and 270 in b. Each neighborhood's early station counted 300 ballots, 150 for X at a's station and 120 at b's. The district reports 200 out-of-zone early ballots, 120 of them for X. Suppose 300 residents of each neighborhood voted early in the zone and 100 outside it.

Two things are unknown. *Movement*: if 30 residents of a voted at b's station and 30 residents of b at a's, the station counts are unchanged, but a's residents may have cast anywhere from 30 more to 30 fewer X ballots than its station shows, a change of 3 percentage points in a's share. *Pooling*: the 120 pooled X ballots may be split so that a receives anywhere from 20 to 100 of them, a further 8-point range. The district's share of X is the same in every case; how it divides between a and b is not. A conventional map, which shows each station's count and drops the pooled ballots, picks one allocation without saying so.

Feasible allocations.

For each neighborhood, the returns record election-day votes, the number of residents removed from the election-day roll, and early ballots counted at each station. They also record candidate totals for each pooled voting mode in the counting jurisdiction. Any allocation of ballots to residents must reproduce these counts and respect the statutory early-voting zones. A conclusion is identified if it holds for every feasible allocation. Linear programming gives the minimum and maximum of each linear quantity of interest. These bounds are sharp when ballot counts are treated as continuous; imposing integer counts changes the reported bounds negligibly (Online Appendix A). This partial-identification approach makes explicit what the data and assumptions establish (Manski 2003; Tamer 2010). Conventional ecological inference concerns unobserved cross-classifications within observed aggregate totals (Duncan and Davis 1953; King 1997; Wakefield 2004; Cho and Manski 2008; Imai et al. 2008; Greiner and Quinn 2009; Jiang et al. 2020). Here, the neighborhood totals of interest are themselves unobserved because reporting intervenes between voting and the published record (Knox et al. 2020; see also Cross and Manski 2002; Molinari and Peski 2006; Ridder and Moffitt 2007). Recent work on monotone ecological inference obtains sharp bounds on individual-level associations from observed area totals by restricting the signs of unobserved associations (Elzayn et al. 2026). Our problem arises one step earlier, because the residential totals are not published, but we follow the same strategy of making the identifying role of each assumption explicit.

3.4 Assumptions and outcomes

The published counts permit extreme allocations, including ones in which neighboring areas' residents account for almost entirely different candidates' out-of-zone ballots. To assess how much the conclusions depend on such possibilities, we impose three restrictions and vary their strength (Manski and Pepper 2018; Rambachan and Roth 2023).

1. *How far did early voters move?* At the lower extreme, early voters moved only as much as the roll forces: when a station counted more ballots than its neighborhood had residents removed from the roll, the difference must have come from neighbors. We call this the *minimal-movement benchmark*; it serves as a lower-movement reference rather than an estimate of actual behavior. For a *mobility-calibrated scenario*, we use mobile-phone trip data (Seoul Living Mobility) to calibrate an upper bound on cross-neighborhood movement: beyond a baseline set by the roll (Online Appendix A.1), the early voters who voted at another neighborhood's station in their zone may number at most the zone's out-of-zone early voters multiplied by the ratio of residents' daytime trips to other neighborhoods of the zone to their trips leaving it. Trips are not votes, so the data set the scale of a cap rather than measure movement; studies that infer turnout from phone visits to polling stations use such

data differently (Harada et al. 2025). We vary the cap by a factor λ and also report unrestricted movement.

2. *How similar are out-of-zone and local-station votes?* We allow the candidate shares of a neighborhood's out-of-zone ballots to differ from those of its early station by at most δ, ten points in the main specification. A version that first shifts the station shares by the district's observed difference between out-of-zone and in-zone early voters yields nearly identical results.
3. *How far can residents' early-vote shares depart from their benchmark?* We benchmark each neighborhood's early vote by its own election-day vote, shifted by the district's observed gap between early and election-day voters. We then allow residents' early ballots, wherever in the zone they were cast, to depart from this benchmark as much as their station's ballots do, plus an additional tolerance s (two percentage points in the main specification). This restriction carries much of the election-day pattern over to early voters: with small s, residents' early ballots inherit their neighborhood's election-day vote, shifted by a district-wide gap. Its role is therefore to ask how far the early-vote gradient could differ from the election-day gradient, which is point-identified (Section 4.2), before a conclusion about all residents changes.

We call the combination of the mobility-calibrated scenario, $\delta = 0.1$ and $s = 0.02$ our *main assumptions*. They are reference points, not validated error bounds. Stations' own ballots depart from the benchmark by a median of about one point, and in the five elections with early voting the conservative share among a district's out-of-zone early voters differs from that among its in-zone early voters by at most 7.7 points (Online Appendix Table C21), so the 10-point tolerance exceeds every district-level gap observed. What δ bounds, however, is the gap for each neighborhood's residents, which is not observed and may vary within a district more than district totals show. In 2026 out-of-zone voters include residents who voted elsewhere in their own district, so the tolerance is arguably more credible then than in presidential elections. We therefore treat as evidence the range of settings under which a conclusion holds: each assumption is indexed by one parameter, so we report how far it can be relaxed before a conclusion changes and what the allocation that first reverses it looks like (Masten and Poirier 2020; Table 3, Panel B; Section 4.5). Box 2 states the accounting identities and the outcome; Online Appendix A gives the assumptions in the same notation.

We study all residents rather than election-day voters alone because election-day voters are a selected and shrinking part of the electorate, 50–66% of ballots in recent elections, whose partisanship differs from early voters' by 10 to 28 points; a claim about a neighborhood's politics concerns everyone who votes there.

Box 2. The accounting identities and the outcome

For each counting jurisdiction (a district, or a constituency in 2024), let A_{ic} be the election-day ballots for candidate c in neighborhood i, K_{jc} the in-zone early ballots counted at the station of neighborhood j, $E_i - D_i$ the residents of i removed from the election-day roll, and P_c and Q_c the jurisdiction totals of out-of-zone early ballots and of at-home and overseas ballots. The unknowns are F_{ijc}, the early ballots of residents of i cast at station j; P_{ic} and Q_{ic}, residents' pooled ballots; and U_i, removed residents who cast no counted ballot. Every allocation must satisfy

$$\textstyle\sum_{j,c} F_{ijc} + \sum_c P_{ic} + \sum_c Q_{ic} + U_i = E_i - D_i, \qquad \sum_i F_{ijc} = K_{jc}, \qquad \sum_i P_{ic} = P_c, \qquad \sum_i Q_{ic} = Q_c,$$

and $F_{ijc} = 0$ when i and j lie in different early-voting zones. Residents' votes for c are $N_{ic} = A_{ic} + \sum_j F_{ijc} + P_{ic} + Q_{ic}$.

The outcome. With A_i the election-day voters of i, let $H_i = A_i + (E_i - D_i)$ and let $I_i = N_{i,\text{inv}}$ be residents' invalid ballots. We analyze the roll-based share $z_{ic} = N_{ic}/H_i$, whereas the usual valid-vote share is $y_{ic} = N_{ic}/(H_i - U_i - I_i)$. Because H_i includes removed residents who cast no counted ballot, z is slightly smaller than y and is not the valid-vote share. We use z because it is linear in the allocation, so its ranges are exact; for y we report bracketing ranges, which lead to the same conclusions (Online Appendix Tables C15 and C18).

The estimands describe registered residents of places, not homeowners, and they are associations, not effects of housing wealth. All results are conditional on the observed returns and price measure. The ranges are identified sets for a finite population of registered residents, not confidence intervals: the returns are a census, so no sample of voters is involved. The point-identified election-day slopes are likewise exact descriptions of the observed neighborhoods; because prices were not assigned to neighborhoods at random, we report permutation p-values and clustered standard errors only as auxiliary diagnostics that rest on further assumptions (Abadie et al. 2020; Online Appendix Table C20). The remaining statistical uncertainty lies in the price measure, which is estimated from transactions; we report how much the ranges move when prices are re-estimated (Online Appendix Table C13). Inference for estimated bounds is developed elsewhere (Imbens and Manski 2004; Stoye 2009; Fang and Santos 2019; Fang et al. 2023).

4. Results

4.1 The association across districts

At the district scale, ballots can be assigned to the residential unit without neighborhood-level allocation assumptions. The 2024 analysis uses 24 units, merging Jung-gu and Seongdong-gu because one constituency spans both; the other elections use 25 districts. For all of Seoul, higher apartment prices are associated with greater conservative support in every election (Figure 2A; Table 3). A doubling of the geometric mean of a district's neighborhood prices is associated with a conservative share 4 percentage points higher in 2017 and 9 to 14 points higher in the other elections. Excluding Gangnam, Seocho and Songpa reduces the slope to about half to two-thirds of its full value in 2022–2026. In 2012 and 2017 it falls to 0.057 and 0.025, less than half its full value, and model-based standard errors would not distinguish these slopes from zero (Online Appendix Table C14). The earlier district-level associations thus rest largely on the three southeastern districts.

4.2 The association within districts

Table 2 reports what can be said about neighborhoods within districts as assumptions are added. Each cell gives the range of the within-district slope, or of the high-minus-low contrast, over all allocations consistent with the data and the stated assumptions; a range that includes zero means that the direction of the relation is not established, which is not the same as showing that there is no relation.

Election-day voters. Election-day ballots are counted in the voter's own voting district, so the slope of the conservative share among election-day voters needs no allocation assumption (Table 2). It is positive in 2012 and in every election from 2022 and close to zero in 2017: 0.097, 0.102, 0.063 and 0.086 in 2022, 2024, 2025 and 2026, and 0.079, 0.078, 0.041 and 0.069 outside Gangnam, Seocho and Songpa (Online

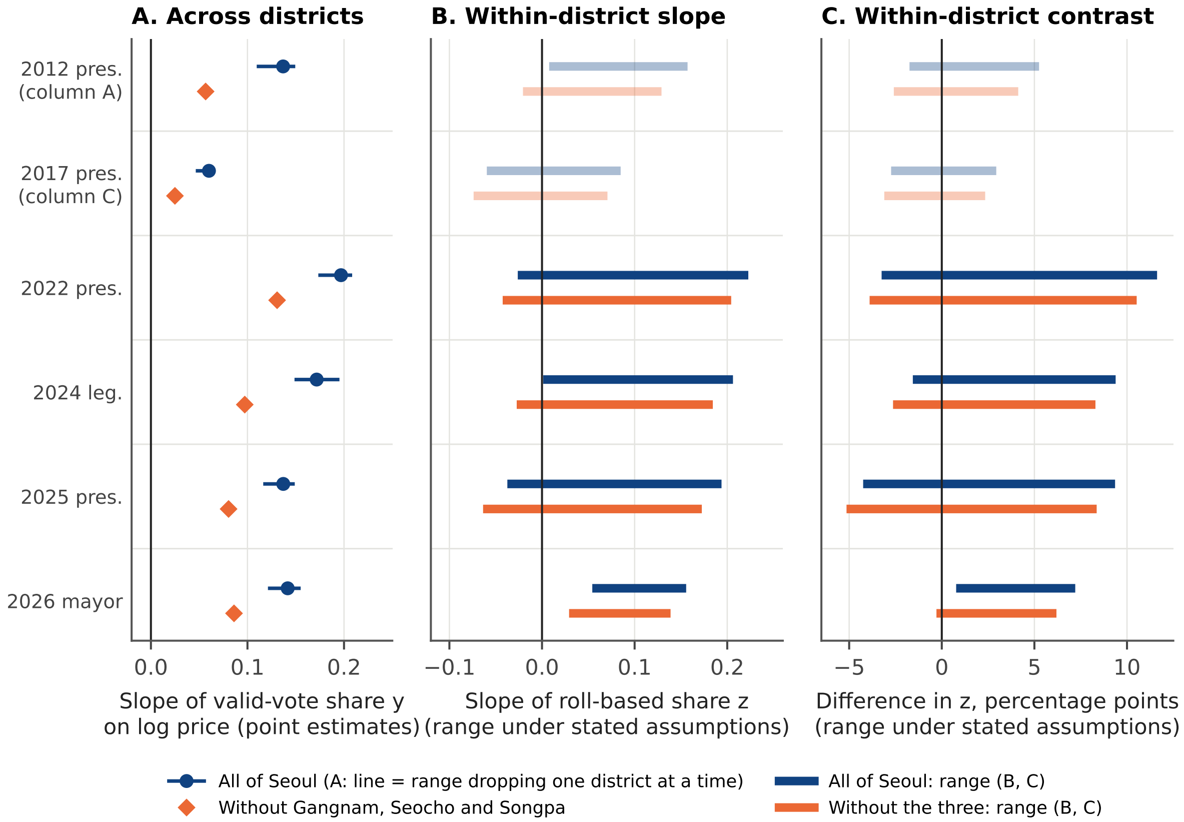


Figure 2: The three spatial claims, for all of Seoul and without Gangnam, Seocho and Songpa. (A) Expensive districts: slope of the conservative valid-vote share on log price across districts; for all of Seoul, the dot is the slope over 25 districts (24 units in 2024) and the line the range of slopes when one district at a time is dropped. (B) Expensive neighborhoods within districts: range of the within-district slope of the roll-based share z consistent with the data under the main assumptions (Table 2, column E). (C) The same for the difference between the more and less expensive halves of each district's neighborhoods, in percentage points. 2012 and 2017, which lack mobility data, use the returns and rolls alone and minimal movement with out-of-zone voters within ten points (columns A and C) and are drawn faint, as they are not comparable with the later elections. Panel A uses valid-vote shares, panels B and C roll-based shares. Ranges are sets of values consistent with the data and assumptions, not confidence intervals.

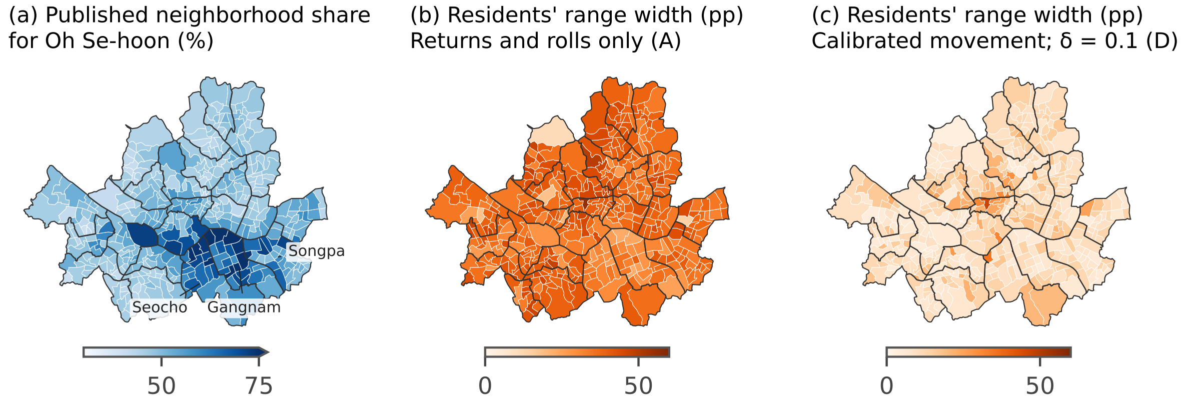
Appendix Table C20). The rest of this section asks whether the same holds for all residents, including the third to half who voted early.

The returns alone. The conventional estimates, computed from the published neighborhood rows, are positive in every election. But with the returns and rolls alone, the within-district relation among all residents could be strongly positive or strongly negative in every election since 2014 (column A); in 2022–2026 the ranges are 7 to 14 times as wide as the conventional estimates themselves, and in 2017, when the conventional estimate is close to zero, far wider still. The contrast between higher- and lower-price halves of each district ranges from about −43 to +47 points in 2022 and from −20 to +25 points in 2026, around conventional estimates of 3 to 4 points. Only in 2012, before nationwide early voting, are the returns informative without assumptions: the slope is positive, although the contrast is not established even then. The returns also say little about individual neighborhoods: with the returns alone, the median neighborhood's conservative share in 2026 is pinned down only within a 39-point range, and the direction of the difference between adjacent neighborhoods is established for none of 445 pairs (Figure 3).

Table 2. Within-district associations under alternative assumptions

Election	Conv.	Election-day voters (exact; valid-vote share y)	(A) Returns and rolls only	(B) Minimal movement	(C) = (B) + out-of-zone voters within 10 points	(D) Mobility-calibrated movement + out-of-zone within 10 points	(E) = (D) + residents' early-vote composition restricted	(F) = (E), price-dependent benchmark
A. Within-district slope								
2012	0.087	0.087	[0.008, 0.157]	[0.008, 0.157]	[0.027, 0.140]	–	–	–
2017	0.014	0.013	**[−0.428, 0.450]**	**[−0.136, 0.162]**	**[−0.060, 0.085]**	–	–	–
2022	0.097	0.097	**[−0.615, 0.704]**	**[−0.100, 0.264]**	[0.014, 0.175]	**[−0.283, 0.448]**	**[−0.026, 0.223]**	**[−0.027, 0.222]**
2024	0.102	0.102	**[−0.553, 0.683]**	**[−0.077, 0.254]**	[0.033, 0.164]	**[−0.185, 0.372]**	[0.001, 0.206]	**[−0.003, 0.200]**
2025	0.077	0.063	**[−0.468, 0.552]**	**[−0.104, 0.232]**	[0.002, 0.149]	**[−0.283, 0.411]**	**[−0.037, 0.194]**	**[−0.062, 0.183]**
2026	0.103	0.086	**[−0.279, 0.439]**	**[−0.046, 0.234]**	[0.072, 0.138]	[0.010, 0.201]	[0.054, 0.156]	[0.052, 0.154]
B. Within-district contrast: higher- minus lower-price groups (percentage points)								
2012	1.9	1.9	**[−1.7, 5.3]**	**[−1.7, 5.3]**	**[−0.4, 4.1]**	–	–	–
2017	0.1	0.1	**[−16.0, 16.1]**	**[−5.1, 5.5]**	**[−2.7, 2.9]**	–	–	–
2022	4.1	4.0	**[−43.4, 46.9]**	**[−8.7, 15.4]**	**[−0.5, 8.5]**	**[−19.2, 26.9]**	**[−3.2, 11.6]**	–
2024	3.8	3.8	**[−30.7, 37.6]**	**[−5.3, 12.4]**	[0.3, 7.0]	**[−11.7, 19.2]**	**[−1.6, 9.4]**	–
2025	2.6	1.8	**[−26.1, 29.7]**	**[−7.2, 11.2]**	**[−1.7, 6.6]**	**[−17.8, 22.5]**	**[−4.2, 9.4]**	–
2026	3.8	3.0	**[−19.6, 25.2]**	**[−6.1, 12.8]**	[1.9, 6.0]	**[−2.4, 10.3]**	[0.8, 7.2]	–

Notes. Bold ranges include zero; their sign is not identified. Panel A: within-district slope of the conservative share among residents (roll-based share z) on log median apartment price per square meter; 0.1 means a neighborhood with prices twice as high has a share about 7 points higher. Within National Assembly constituencies in 2024 the ranges are [−0.729, 0.816] (A), [0.004, 0.166] (C), [−0.269, 0.427] (D) and [−0.035, 0.221] (E). Panel B: mean share of each district's neighborhoods priced above its median minus that of the others, averaged over districts weighted by neighborhoods with price data, in percentage points. Conv.: published neighborhood rows, pooled ballots dropped. Election-day voters (exact): the valid-vote share among residents who voted on election day, point-identified (Online Appendix Table C20). (A) every allocation consistent with the counts, rolls and statutory zones; (B) minimal movement; (C) adds out-of-zone ballots within 10 points of the station; (D) replaces (B) with mobility-calibrated movement; (E) adds residents' early ballots within their station's departure plus 2 points of the benchmark; (F) price-dependent benchmark. Definitions in Section 3.4 and Online Appendix A.


(a) Published neighborhood share for Oh Se-hoon (%)

(b) Residents' range width (pp) Returns and rolls only (A)

(c) Residents' range width (pp) Calibrated movement; δ = 0.1 (D)

Figure 3: 2026 Seoul mayoral election. (a) Conventional neighborhood share for Oh Se-hoon (published station counts, pooled ballots dropped), in percent. (b) Width, in percentage points, of the range of each neighborhood's residents' share consistent with the returns alone (Table 2, column A). (c) The same under mobility-calibrated movement with out-of-zone voters within ten points of their station (Table 2, column D). Panels (b) and (c) share one scale; pp denotes percentage points. Lines are district boundaries; neighborhood boundaries from vuski (2026).

Adding assumptions. The minimal-movement benchmark (column B) establishes no direction after 2012. Adding that out-of-zone voters resemble their station within ten points (column C) yields positive slopes in every election except 2017, but only because the benchmark confines most early voters to their own neighborhood. With mobility-calibrated movement instead (column D), the relation is again undetermined in 2022, 2024 and 2025 and narrowly positive in 2026. Adding the restriction on residents' early-vote composition (column E) makes the 2026 conclusion robust and yields a fragile positive slope in 2024, lost within constituencies or with a price-dependent benchmark (column F); 2022 and 2025 remain undetermined.

Gradient and contrast. The contrast concerns a different summary of the association. Under mobility-calibrated movement, only in 2026, and only with the third assumption, are the higher-price halves of districts established to vote more conservatively than the lower-price halves: by between 0.8 and 7.2 points, around a conventional estimate of 3.8 (Figure 2C). Within districts, then, the evidence for a price gradient is stronger than the evidence that higher-price and lower-price neighborhoods differ as groups.

A published allocation. The research note cited in the introduction (Lee 2026) allocated the pooled ballots explicitly. Re-implementing its published description reproduces its reported correlations (0.842 and 0.709) and slope (1.88 points per 10 million won per pyeong) to their reported precision, and that slope lies within the range identified under our main assumptions ([1.16, 2.54] in the same units). The allocation itself is not feasible: in 47 of 155 constituencies it assigns more pooled ballots than the roll leaves, by a median of 16 voters (Online Appendix D). A single allocation cannot show, however, how far its conclusion depends on the allocation rule: the same procedure would return one slope for 2022 or 2025, where allocations satisfying our restrictions give slopes of both signs.

4.3 Beyond Gangnam, Seocho and Songpa

To assess whether the 2026 within-district result depends on Gangnam, Seocho and Songpa, we exclude their neighborhoods from the estimands. We recompute the regression and contrast weights for the 357 neighborhoods with price data among the 360 in the remaining 22 districts while retaining all accounting constraints (Table 3 and the orange bars in Figure 2).

Among election-day voters the slope outside the three districts is positive in 2022–2026 (Online Appendix Table C20). Among all residents, under the main assumptions, the 2026 within-district slope remains positive after the exclusion: its range is 0.029 to 0.139, compared with 0.054 to 0.156 for all of Seoul. In magnitude, a neighborhood whose apartment prices are twice those of another in the same district has a conservative roll-based share 2.0 to 9.6 points higher in the fitted relation; this describes the regression slope, not the effect of a price change or a difference found between every pair of neighborhoods. The median-split contrast ranges from −0.3 to 6.2 percentage points after the exclusion and the top-minus-bottom-tercile contrast from 0.1 to 7.9 (Online Appendix Table C23), so the group comparisons are borderline. Valid-vote shares yield the same conclusions about signs (Online Appendix Table C18). Excluding the three districts also removes the positive lower bound for the 2024 slope under the main assumptions and for the 2012 slope based on returns and rolls alone (Table 3).

The positive slope outside the three districts requires the restriction on residents' early-vote composition: under mobility-calibrated movement and the ten-point tolerance alone (column D), the slope without them ranges from −0.020 to 0.188. Holding movement and the tolerance fixed, it stays positive as long as residents' early ballots depart from their benchmark by no more than about 0.103 beyond their station's departure, about five times the main setting's 0.02; with unrestricted movement the threshold is 0.067. The result is more sensitive to the out-of-zone tolerance: with a 20-percentage-point tolerance

Table 3. Associations with and without Gangnam, Seocho and Songpa

Election	Across districts: all	Across districts: without the three	Within districts, slope: all	Within districts, slope: without the three	Higher-minus lower-price half (points): all	Higher-minus lower-price half (points): without the three
A. Across districts: point values; within districts: ranges						
2012 (A)	0.137	0.057	[0.008, 0.157]	**[−0.021, 0.129]**	**[−1.7, 5.3]**	**[−2.6, 4.1]**
2017 (C)	0.060	0.025	**[−0.060, 0.085]**	**[−0.074, 0.071]**	**[−2.7, 2.9]**	**[−3.1, 2.3]**
2022	0.197	0.131	**[−0.026, 0.223]**	**[−0.042, 0.204]**	**[−3.2, 11.6]**	**[−3.9, 10.5]**
2024	0.172	0.097	[0.001, 0.206]	**[−0.027, 0.184]**	**[−1.6, 9.4]**	**[−2.6, 8.3]**
2025	0.137	0.080	**[−0.037, 0.194]**	**[−0.064, 0.173]**	**[−4.2, 9.4]**	**[−5.1, 8.4]**
2026	0.142	0.086	[0.054, 0.156]	[0.029, 0.139]	[0.8, 7.2]	**[−0.3, 6.2]**
B. 2026 sensitivity thresholds s^* (not slopes): largest additional departure at which the lower end stays positive; main setting $s = 0.02$						
Mobility-calibrated, ±10	–	–	∞	0.103	0.049	0.010
Mobility-calibrated, ±20	–	–	0.087	0.001	none	none
Unrestricted movement, ±10	–	–	0.122	0.067	0.019	none
Unrestricted movement, ±20	–	–	0.006	none	none	none

Notes. Bold ranges include zero; their sign is not identified. Across districts: slope of the valid-vote share on log price across districts (point values, no allocation assumption). Within districts: ranges in z under the main assumptions (Table 2, column E); 2012 under column A and 2017 under column C, as those elections lack mobility data. Without the three: Gangnam, Seocho and Songpa dropped from the regression and the contrast, all accounting constraints kept. Panel B: s^* with movement and the out-of-zone tolerance (±10 or ±20 points) held fixed; a relation is established at the main setting when s^* exceeds 0.02. ∞: positive however much residents' early ballots depart; none: not positive for any s (Online Appendix Table C19).

and mobility-calibrated movement, the slope remains positive only up to an additional tolerance of 0.001, below the main setting (Table 3, Panel B; Online Appendix Table C19). The conclusion beyond Gangnam is therefore less robust than the conclusion for all of Seoul, which survives a 20-point tolerance up to $s = 0.087$.

Conditional on these assumptions, the 2026 slope does not support an account confined to Gangnam, Seocho and Songpa. It does not establish a positive slope in each of the other 22 districts or identify a causal effect of housing prices, and it is compatible with place identity operating beyond Gangnam (Bae and Joo 2020).

4.4 Information supplied by early-voting zones

The elections differ sharply in what they reveal, and one reason is administrative. In presidential elections the early-voting zone was the whole district; in 2026 it was the district-council constituency, about three neighborhoods. Because the 2026 returns can be analyzed with and without the zone rule, the role of the rule can be isolated in the same data (Table 4).

With the uniform movement cap held fixed, the within-district slope ranges from −0.039 to 0.239 when early ballots may move throughout the district, compared with 0.014 to 0.195 under the statutory zones. Imposing the zones therefore changes whether a positive slope is identified under these assumptions. The mobility-calibrated comparison also changes the sign conclusion ($[-0.191, 0.373]$ versus $[0.010, 0.201]$),

Table 4. Within-district slopes with and without statutory zones, 2026

Assumptions	Zones ignored: in-zone early ballots may be cast anywhere in the district	Statutory zones: only within the voter's council constituency
Returns and rolls only	**[−0.378, 0.519]**	**[−0.279, 0.439]**
Minimal movement, out-of-zone within 10 points	[0.072, 0.139]	[0.072, 0.138]
Uniform movement (η = 0.1), out-of-zone within 10 points	**[−0.039, 0.239]**	[0.014, 0.195]
Mobility-calibrated movement, out-of-zone within 10 points (D)	**[−0.191, 0.373]**	[0.010, 0.201]
Main assumptions (E)	[0.034, 0.174]	[0.054, 0.156]

Notes. Bold ranges include zero; their sign is not identified. Roll-based share z. The zones are the 155 district-council constituencies, about three neighborhoods each. Rows with uniform movement hold the amount of movement fixed, so the comparison isolates the zone rule; the mobility-calibrated rows recalibrate movement to the zone definition. Adding the council contest's published headcounts changes the statutory-zone ranges by at most 0.003 (Online Appendix Table C2).

although movement is recalibrated to each zone definition. Under the main assumptions, both ranges are positive and the zones narrow the range by about a quarter; there, the composition restrictions rather than the zones secure the sign. The comparison holds the 2026 votes and pooled-ballot structure fixed and changes only where in-zone ballots may be allocated. It is not a counterfactual election under presidential rules, in which today's cross-constituency voters would have been in-zone movers, and it does not estimate how much of the difference between 2025 and 2026 is administrative; for the same reason, δ constrains a different population in 2026 than in presidential elections. The comparison shows only that, within the same returns, finer zones carry information that can decide the sign under weaker assumptions. Cross-election differences also reflect candidates, issues and the partisanship of early voting, so the clearer result of 2026 does not by itself show that Seoul's housing divide deepened. Comparing political divisions across elections requires first establishing that each election's counting rules reveal the quantity being compared.

4.5 Sensitivity and robustness

Figure 4 maps how far the movement and composition assumptions can be relaxed: for each amount of movement and each out-of-zone tolerance, the largest additional departure s^* at which the slope stays positive. For all of Seoul in 2026, with a 10-point tolerance the slope stays positive at the main setting for every amount of movement, including unrestricted movement ($s^* = 0.122$). With a 20-point tolerance, s^* is 0.087 under calibrated and 0.006 under unrestricted movement (Online Appendix Tables C4 and C19). The 2026 conclusion therefore depends mainly on the out-of-zone tolerance, not on the mobile-phone calibration; at the main s it holds up to $\delta = 0.26$ for all of Seoul and $\delta = 0.18$ outside the three districts (Online Appendix Table C25). In 2024 the region is a narrow strip ($s^* = 0.021$ at the main movement and tolerance); in 2022 and 2025 there is none.

Price-aligned departures. A reversal requires residents' early votes in higher-price neighborhoods to be systematically less conservative than their neighborhoods' election-day votes imply, but the symmetric restrictions let each neighborhood depart independently. We therefore ask what a different restriction would imply: within each counting jurisdiction, the slope of residents' early-vote departures (in-zone and out-of-zone, as a share of the roll) on log price may not exceed κ in absolute value (Online Appendix A.1 and Table C22). With the out-of-zone tolerance removed, the slope among all residents stays positive for κ up to 0.042, 0.038, 0.032 and 0.095 in 2022, 2024, 2025 and 2026, and up to 0.028, almost zero, 0.011 and

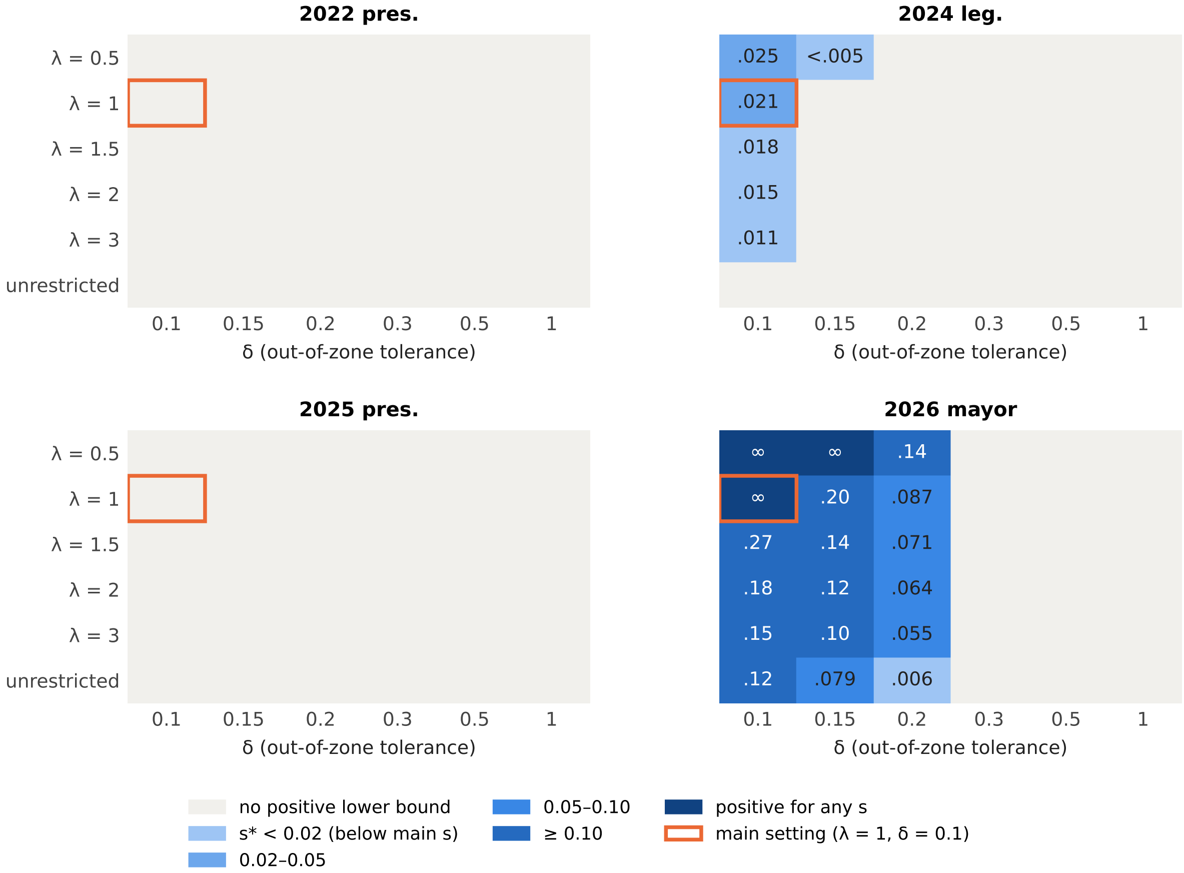


Figure 4: Where the within-district slope is positive. Each cell gives s^*, the largest additional departure of residents' early ballots from their neighborhood benchmark, beyond their station's own departure, at which the lower end of the range stays positive; amounts of movement are multiples λ of the mobility-calibrated scenario or unrestricted, and δ is the tolerance for out-of-zone voters. The main setting ($\lambda = 1$, $\delta = 0.1$, $s = 0.02$) is outlined: the slope is positive there if s^* exceeds 0.02. "∞": positive however much residents' early ballots depart; gray: no positive lower bound for any $s \geq 0$ among the settings examined. All of Seoul; numbers in Online Appendix Table C4, and selected settings without Gangnam, Seocho and Songpa in Table C19.

0.075 outside the three districts. Two observable patterns give a scale, though neither measures residents' unobserved departures: the same slope for in-zone station ballots, which also count non-residents, is at most 0.016, 0.047, 0.026 and 0.013 in absolute value across jurisdictions, and the slope implied if early-vote gaps varied with price within counting jurisdictions as they do across them (districts; constituencies in 2024) is 0.007, 0.008, 0.021 and 0.006. For all of Seoul κ^* exceeds both scales in 2022, 2025 and 2026 and the between-jurisdiction one in 2024; outside the three districts it exceeds both in 2022 and 2026 and neither in 2024 and 2025. Bounding only the alignment of out-of-zone departures orders the elections similarly (Table C25). These results identify the additional assumption that would settle the undetermined signs; they do not show that it holds.

Table 5 describes the allocations that overturn each result. In 2022 and 2025 the sign is lost as soon as residents may depart from their benchmark as much as their own station does. The reversing allocation makes residents who voted at another station 12–24 points more conservative than those who used their own, in every price tercile (Online Appendix Table C5). In 2022 it requires about 17,000 residents' early ballots to depart from their benchmark, 95% of them in the tenth of neighborhoods with the most regression weight, and fewer departures in total than the published stations already show (Table C6).

Restricting movers to within five or ten points of their home station does not help, even together with the composition restriction, because a reversal can then use pooled ballots instead (Tables C7 and C25). In 2026 no reversal exists under calibrated movement unless out-of-zone voters may differ from their station by more than ten points; with a 20-point tolerance, residents' early ballots must depart from their benchmark by about 9 points more than their station's ballots do.

Table 5. Assumptions and allocations that permit a non-positive slope

Election	Out-of-zone voters within (points)	Largest additional tolerance s with a positive slope	First reversing allocation: movers minus stayers, conservative share (points)	Fewest residents' early ballots that must depart to reverse
2022	10	none	12 to 21	16,685
2022	20	none	14 to 18	15,340
2024	10	0.021	11 to 17	32,177
2024	20	none	9 to 14	30,103
2025	10	none	20 to 24	13,539
2025	20	none	21 to 23	12,512
2026	10	∞	no reversal	no reversal
2026	20	0.087	−2 to 24	69,384

Notes. Mobility-calibrated movement. s: residents' early ballots may depart from their neighborhood benchmark (election-day share shifted by the district's early-vote gap) by their station's departure plus s; ∞: positive however much they depart; none: not positive even when they depart only as much as their station. First reversing allocation: the one with the lowest slope at s just above that value; movers voted at another station of their zone, stayers at their own; range over price terciles. Last column: among allocations with a non-positive slope, the fewest residents' conservative early ballots departing from the benchmark (out-of-zone departures and concentration in Online Appendix Table C6). Each reversal is minimal only in its own metric, with other assumptions fixed.

Alternative specifications.

The 2026 conclusion under the main assumptions holds with regressions weighted by H_i, mean prices, without the 43 neighborhoods with fewer than 30 transactions, and without the 5% or 10% most expensive neighborhoods, but it depends on the most expensive neighborhoods: without the top 10% the lower bound falls from 0.054 to 0.008, and with $s = 0.05$ it turns negative (Online Appendix Table C10); resampling transactions moves the bounds by a few thousandths (Table C13). A benchmark that varies with price within districts, motivated by larger early-vote gaps in higher-price districts in 2024–2026, changes the 2026 result only slightly (Table 2, column F), as does shifting the out-of-zone benchmark by the observed out-of-zone gap (Table C9). Valid-vote shares y lead to the same conclusions, except that column C does not establish the 2025 slope for them (Table C15), and the council contest's out-of-zone headcounts change the 2026 ranges by at most 0.003 (Table C2). Holding neighborhood age composition fixed leaves the election-day slopes almost unchanged and the 2026 range among all residents positive ($[0.046, 0.160]$; $[0.024, 0.148]$ outside the three districts), but removes the fragile 2024 lower bound (Table C23).

5. Reporting reforms

The uncertainty is partly a product of what is published, and it could be reduced without publishing any individual vote. Out-of-zone early ballots already arrive at district commissions in individually tracked envelopes. *Residence reporting* would sort them by the voter's registered neighborhood before opening and publish aggregate counts by neighborhood. At-home and overseas ballots, which also arrive by mail, could in principle be reported the same way. *Headcount reporting* would publish how many residents of each neighborhood voted at each early station, with no vote choice.

Because the true allocation is unknown, we evaluate each change by the uncertainty it is guaranteed to remove: the largest range that could remain whatever the true allocation, computed as the largest difference in the within-district slope between two allocations that the new report could not tell apart. Table 6 reports, under the main assumptions, the current range and the largest range that could remain after each report.

Table 6. Guaranteed reductions in uncertainty under residence reporting

Election	Current range width	Proposal: out-of-zone early ballots by neighborhood, share removed	Proposal: maximum remaining width	All pooled modes by neighborhood, share removed	All pooled modes: maximum remaining width
2022	0.25	27%	0.18	57%	0.11
2024	0.20	30%	0.14	49%	0.11
2025	0.23	32%	0.16	61%	0.09
2026	0.10	57%	0.04	69%	0.03

Notes. All columns describe the width of the within-district slope range or the guaranteed fraction of that width removed. Main assumptions: Table 2, column E. Proposal: out-of-zone early ballots reported as aggregate counts by the voter's registered neighborhood, at-home and overseas ballots kept pooled. All pooled modes: out-of-zone early, at-home and overseas ballots by neighborhood. Maximum remaining width: the largest difference in the within-district slope between two allocations that the report could not tell apart, which bounds the range left whatever the true allocation. Other assumptions and headcount reporting: Online Appendix Table C12.

Residence reporting of out-of-zone early ballots, with at-home and overseas ballots kept pooled to protect small cells, is the version we recommend. Under the main assumptions it is guaranteed to remove 27%, 30%, 32% and 57% of the current range in 2022, 2024, 2025 and 2026; reporting all three pooled modes would remove 49–69% (Online Appendix Table C12). Neither version guarantees that a currently undetermined direction is revealed: in 2022, two allocations with slopes of −0.026 and +0.026 produce the same residence report (Table C16). Headcount reporting alone reveals little about how movers voted, but combined with reporting all pooled modes it removes 63–72% of the range that remains without any movement assumption (Table C12). Better reporting would narrow the question of how far Seoul's housing politics reaches inside districts without answering it for every election. Counts by registered neighborhood could be released after the official count with small cells suppressed, leaving ballot secrecy and the timing of results unchanged.

Implementation would require sorting out-of-zone envelopes into about seventeen neighborhood groups per district and suppressing small cells. The statistics would also show neighborhood differences between residents' early and election-day votes, which have figured in fraud claims since 2020; they would help researchers relate these gaps to neighborhood composition but would not by themselves settle disputes about them. Headcount reporting is a practical first step: it measures movement without publishing new candidate totals.

6. Discussion and conclusion

The findings bear on three aspects of housing politics in Seoul. In space, the price–vote association recurs across districts in all six elections, as Han (2024) documented for districts, although in 2012 and 2017 it rested largely on Gangnam, Seocho and Songpa; within districts, a pooled gradient extends beyond those three districts, although the evidence does not show a relation in each district. Among voters, the within-district gradient is exact for those who voted on election day and appears in 2012 and in every election from 2022, including outside the three districts. Extending it to all residents, including early voters who now vote very differently, is warranted in 2026 under the main assumptions and narrowly in 2024; in 2022 and 2025 it requires a further, unverifiable restriction on how early-vote departures align with prices. In election administration, the 2026 counting rules reveal more about where votes come from, so a clearer within-district result in 2026 says more about the evidence than about any change in the housing divide. Comparisons of higher- and lower-price groups outside the three districts are borderline, and the evidence does not choose between economic interests, residential sorting and local context, including place identity (Bae and Joo 2020).

These distinctions matter for how neighborhood studies are read. Analyses of published dong returns, including Kim and Kang's (2022) study of price levels and Yun and Jun's (2024) study of price changes, face the attribution problem unless they account for pooled ballots and early voters' movement, although our analysis does not test their estimates directly. Explicit allocations such as Lee's (2026) make their assumptions visible but yield a single slope even where the sign is undetermined. Neighborhood analyses should report the treatment of unattributed ballots, show which conclusions depend on that treatment, and report the election-day relation, which needs none.

The attribution problem also arises where ballots cast away from a voter's assigned polling place are reported at a larger scale. German precinct results omit postal ballots, which were close to half of all ballots in 2021; Australian booth results omit declaration votes (Online Appendix B.5); and many American states report absentee and early ballots only at the county level, requiring precinct datasets to allocate them by assumption (Amos et al. 2017; Baltz et al. 2022). Other systems trade detail against attribution differently (Online Appendix Table B2). Taiwan requires voters to vote at the station of their household registration, so station returns are residence-based. Japan mixes all ballots, including early ballots, at the counting district (開票区), usually a whole municipality or ward, and reports nothing below it: its returns contain no misattribution within that unit but cannot speak to neighborhoods at all. Korea lies between these, publishing neighborhood rows that mix residents' and neighbors' early ballots and omit pooled ones. For the spatial claims examined here, such fine but mixed returns support conclusions that exact but coarse returns could not address, provided that their mixing is bounded explicitly. Applying our approach elsewhere requires small-area returns, candidate totals for pooled ballots, and compatible information on registered residents' use of voting modes. Whether a particular system publishes enough information is an empirical question; where it does, the method can assess the spatial claims its returns support.

Several limitations remain. Mobile-phone trips calibrate possible voting movements rather than observe them, and the composition restrictions concern unobserved ballots. Excluding Gangnam, Seocho and Songpa is one test of geographic reach among many possible. Apartment prices measure neither ownership nor household wealth, and neighborhood age composition, which is related both to prices and to conservative voting, is held fixed only in a robustness check that uses 2026 age data for every election. Constituency results in 2024 combine party and candidate effects, and 2026, the only election with a robust positive result among all residents under the main assumptions, was a mayoral election with a popular incumbent; Table 4 isolates the information supplied by its zones, but not how far the result reflects that election's candidates and turnout. Finally, the estimands concern registered residence, which may differ from actual residence for students, soldiers and residents abroad.

Claims of a housing cleavage therefore need to specify their geographic scope. A district-level association, a slope within districts and separation between price groups provide different evidence, and comparisons across elections need to ask what each counting rule reveals. Reporting out-of-zone ballots by registered neighborhood would make these distinctions easier to evaluate and reduce the role of allocation assumptions in accounts of Seoul's political geography.

Data availability

Replication files, including all data, code and a single command that regenerates every table and figure, will be deposited in a public repository upon acceptance; an anonymized copy is provided for review.

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